Who should we call first?

A stroke-risk model to prioritise prevention outreach

[Your name] · [Date]

A ranking problem: put the highest-risk patients first

1. **Explore & clean**
   Understand the data, find and fix quality issues
2. **Baseline model**
   Simple, interpretable logistic regression
3. **Stronger models**
   Random forest and gradient boosting, compared fairly
4. **Decide**
   Choose a call volume and estimate the impact
5. **Recommend**
   Advice for the outreach team, plus next steps

How success is measured

- Coordinators can only call a small share of patients each month
- So what matters is how many real high-risk patients sit near the top of the list
- **Assumption: outreach targets adults (18+)**

Strokes are rare, and the data needed cleaning

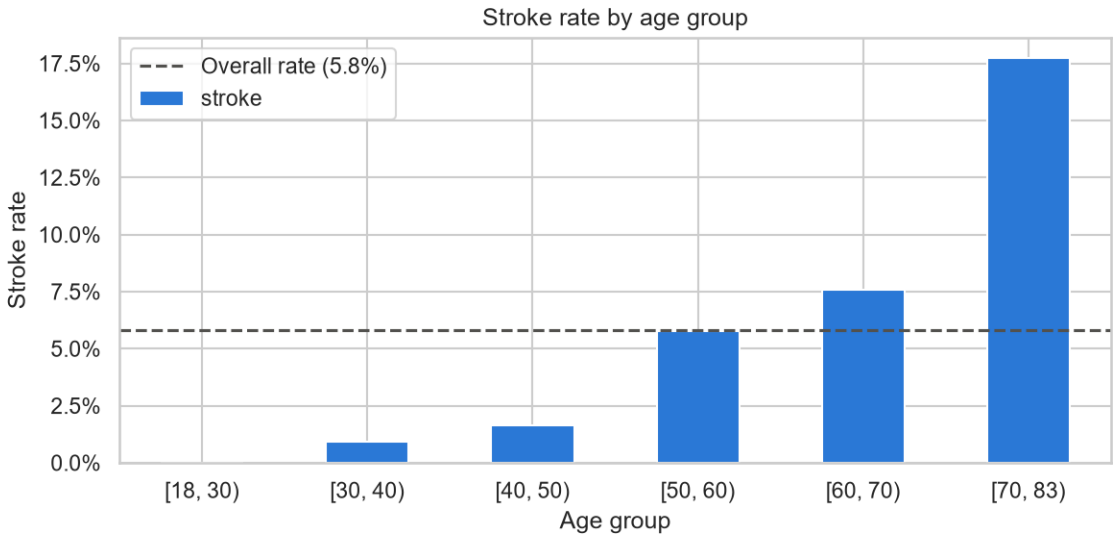
4.9%

of 5,110 patients had a stroke

An "always no" model would be 95% accurate and find nobody, so accuracy is the wrong measure.

Issue	Decision
BMI missing (201)	Keep + 'missing' flag; fill with training median
Smoking "Unknown" (30%)	Keep as a category (mostly children)
BMI over 60 (13)	Cap at 60, likely entry errors
Gender "Other" (1)	Remove; can't learn from one case
Under-18s (856, 2 strokes)	Exclude; outside adult outreach
Duplicates	Checked: none found

Age is the strongest driver of stroke risk



~18%

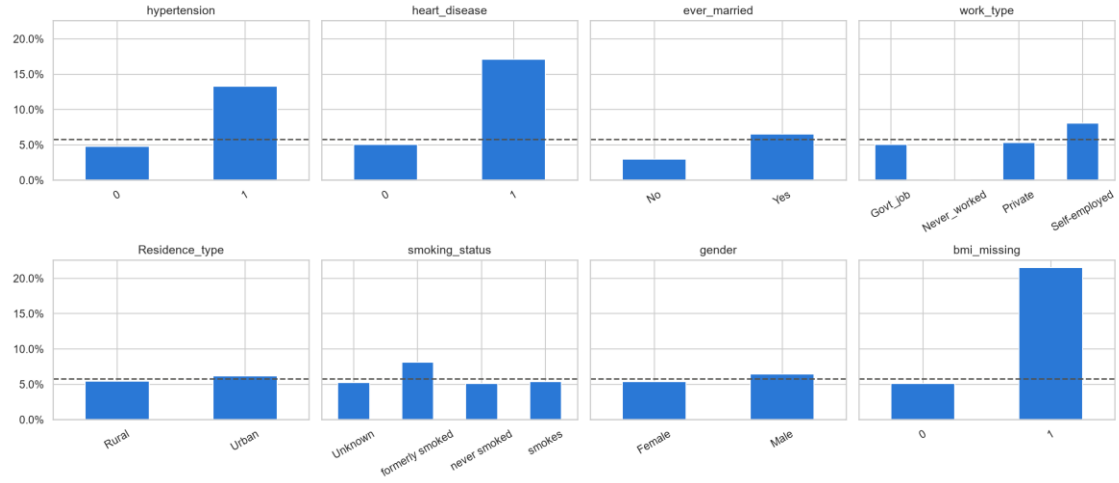
stroke rate for patients aged 70+

3×

the adult average of 5.8%

73% of all strokes are in patients aged 60+.

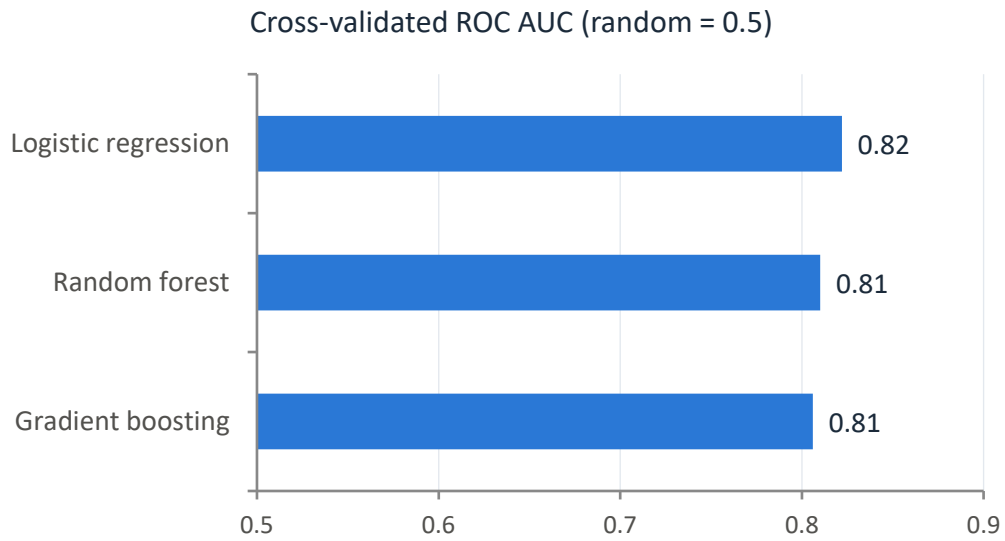
Health conditions matter; some patterns are just age



Dashed line = overall stroke rate

- **Hypertension, heart disease, high glucose: 2–3× the rate**
- **Missing BMI: 20% stroke rate vs 4%**
- Gender, residence, smoking: little difference
- "Married" and "self-employed" look risky only because they're older (confounding)

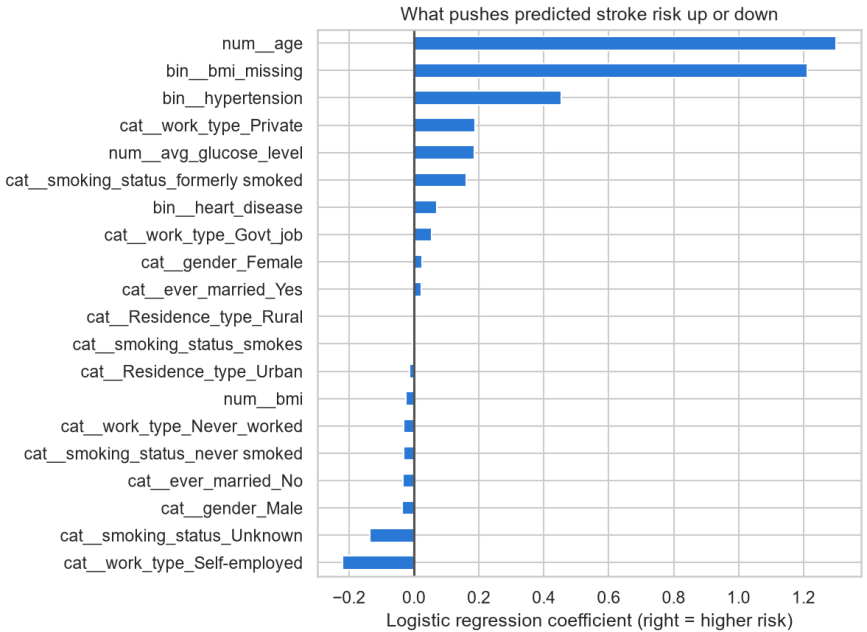
More complex models didn't beat the simple one



PR AUC: 0.23 / 0.22 / 0.23 (random = 0.06)

- **Differences are smaller than the noise between folds: a tie**
- Compared with 5-fold cross-validation; test set kept sealed
- Default random forest overfit: 1.00 on training vs 0.77 on new patients
- Three different model types hitting the same ceiling suggests the limit is the data

Recommended model: logistic regression



0.82

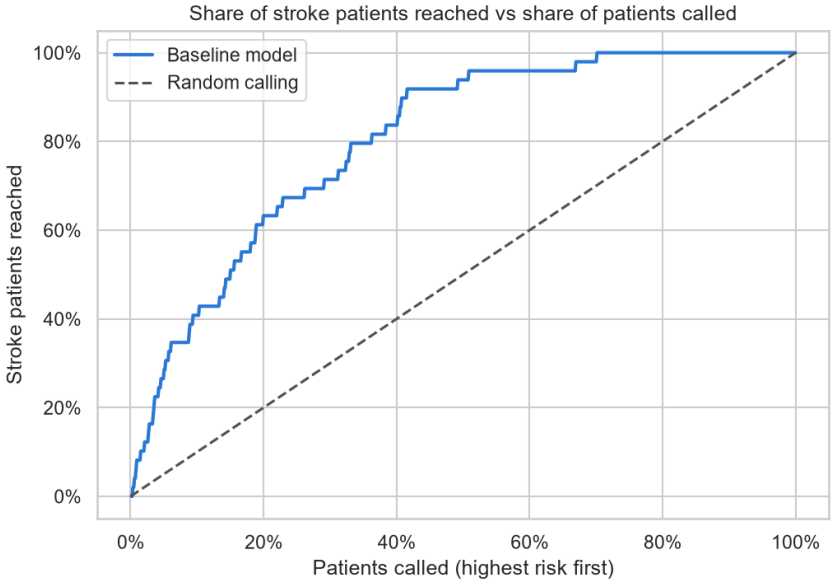
ROC AUC on unseen test patients

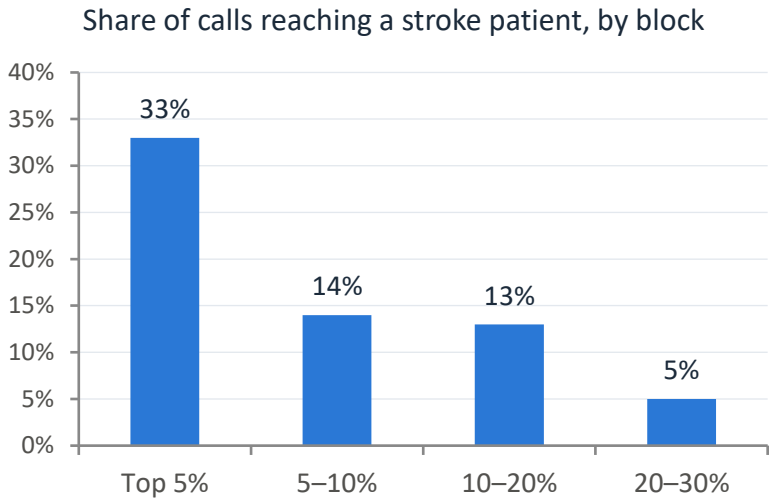
4×

better than random (PR AUC 0.23 vs 0.06)

- Top drivers: age, missing BMI, hypertension, glucose
- Coordinators can see why a patient is flagged
- Simple to run monthly and to maintain

Each extra call finds fewer stroke patients





Beyond the top ~20%, calls hit at the random rate (5.8%).

Call from the top down, up to about 20% of patients

Call the top…	Stroke patients reached	Hit rate	vs random
5%	~29%	1 in 3	~6×
10%	**~41%**	**1 in 4**	**~4×**
20%	~63%	1 in 5–6	~3×
30%	~71%	1 in 7	~2×

Based on 49 test-set strokes, so treat these as approximate.

The recommendation

- Score all adult patients monthly
- Call from the top down as far as capacity allows
- Stop around the top 20%, where the model stops adding value
- If capacity changes, move the cutoff. No retraining needed

Risks to check before launch

1

Possible leakage

BMI may be missing because of a stroke admission. Tested: without the flag, scores barely change (CV PR AUC 0.23 → 0.20).

2

Timing of the data

The target is "had a stroke". We don't know whether features were recorded before or after it.

3

Risk ≠ prevention

The model finds high-risk patients. It doesn't prove that outreach prevents strokes.

Parking lot: what I'd do with more time

Richer clinical data

BP readings, atrial fibrillation, cholesterol, prior mini-strokes

Confirm timing

Ask the data owner how BMI and other features are recorded

Predict the future

Rebuild the target as "stroke in the next X years"

Fairness check

Compare reach by gender, residence and age band

Calibration

Check risk %s before showing them to clinicians

Pilot & monitor

Measure whether outreach reduces strokes; watch for drift